

Econ 30065: Microeconomics

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Game Theory: One-stage games

What is Game Theory

So far, no strategic interaction:

- Individual utility depends only on the bundle consumed by the single consumer
- In perfect competition a firm is price taker
 - ➔ it cares only about its Q
- In Oligopoly (few sellers, many buyers) we see strategic interaction (Q or P chosen by a competitor influences a firm's decision, and thus profits)

What is Game Theory

- Game Theory is used to study situations of *strategic interaction*
 - Utility of a consumer (or profit of a firm) depends not only on his action, but also on the actions taken by other agents
 - When deciding, think also at what actions others may be taking
- Examples:
 - Assume Anna prefers to go to the theater with Michele to go alone (she gets higher utility)
 - ➔ Situation of strategic interaction because utility of Anna depends also on the decision of Michele of joining her
 - Assume Coke and Pepsi are the only 2 firms selling cola on the Japanese market (they are oligopolists). The pricing decision of Pepsi influences the profits of Coke, because if Pepsi sets a price lower than Coke, it will get all the demand
 - ➔ Situation of strategic interaction because profits of Coke depends also on the pricing decision of Pepsi

What is Game Theory

Since utility (or profits) is determined also by others' actions, an agent chooses his optimal action depending on the actions taken by other agents

- In the example of Ann and Michele: if the utility of Ann from going with Michele to the teather is higher than going alone, and staying home is better than going alone, if Michele joins her, she prefers to go, otherwise she prefers to stay home!

What is a game?

- **Game:** a situation of strategic interaction
 - Situation in which a number of individuals make decisions, and each cares both about his own choice and about others' choices:
 - Military planning, dating, auctions, negotiation, oligopoly
- Two types of games:
 - ***One-stage game:*** each participant makes all choices before observing any choice of any other player (simultaneous moves)
 - Rock-Paper-Scissors
 - ***Multiple-stage game:*** at least one participant observes a choice by another participant before making some decision of her own (next time)
 - Poker

One-stage games

Description of the game:

- Set of players

 - Who is playing

- Set of actions

 - What actions are available

- Payoffs

 - How much do agents get once choices are made

Example continued

Suppose Anna and Michele cannot communicate in advance. They will find it out whether the other person decided to go to the theater only when they get to the theater. Moreover suppose that:

Anna's utility from going with Michele to the theater is 4, while Michele's utility is 5.

Anna's utility from asking to Michele to join but going to the theater alone is 2, while Michele's utility is 8.

Anna's utility from letting Michele go alone to the theater is 1, while Michele's utility is 1.

Anna's utility from both not going to the theater is 3, while Michele's utility is 4.

One-stage games

- If number of actions and number of players is finite, represent it with a payoffs-matrix
- This is called the normal form of the game:

		Michele	
		Theater	No theater
Anna	Theater	4,5	2,8
	No theater	1,1	3,4

- Given this structure, we want to find what action each player chooses

Strategic thinking

How do players choose an action in simultaneous moves games?

- Ann cannot see Michele's chosen action before choosing her action (and the same for Michele)
- Since her payoff depends also on what Michele will do, she needs to strategically think at what Michele could choose (put herself in Michele's shoes)
- Ann realizes that what Michele will do depends on what he thinks she will do, and so on...
- The strategic thinking is a never ending story...

Strategic thinking

Good news:

- There are some cases in which all that reasoning is not necessary
 - Dominant actions
- Other cases in which it is easy to guess what the other player will not do for sure
 - Dominated actions

Best response

- **Best response:** an action that provides player with the highest possible payoff, assuming other players behave in a specified way:

Here what is the best response of Anna if we assume Michele always chooses the theater?

➔ THEATER

What about the other cases?

		Michele	
		Theater	no theater
Anna	theater	4,5	2,8
	No theater	1,1	3,4

Dominant action

- **Dominant action:** a player's only best response, regardless of other players' choices
 - When a player has a dominant action, she does not need to think about what other players will do:

Here no matter what Anna does, the best response of Michele is always no theater!

		Michele	
		Theater	no theater
Anna	theater	4,5	2,8
	No theater	1,1	3,4

Anna can easily guess that Michele will not join her, and thus she will prefer not to ask and stay home!

- Not all games have dominant actions!

Dominated actions

- If a player has a dominant action, and he is rational, he will always play it and the opponent predicts it
- Even if the action to choose is not obvious, can sometimes identify actions a player will **not** choose: *dominated actions*
- An action is ***dominated*** if there is some other action that yields a strictly higher payoff regardless of others' choices
 - No rational player will select a dominated action

Thinking Strategically: Iterative Deletion of Dominated action

- **Iterative deletion of dominated actions:** the process of removing the dominated actions from a game, resulting in a simplified game
 1. Remove the dominated actions from a game
 2. Inspect the simplified game to determine whether it contains any (new) dominated actions. If it does, remove them
 3. Repeat this process until there are no more dominated actions left to remove

Iterative deletion of dominated actions

- Sometimes this procedure leads to an answer of which actions the players will choose (examples above)
- However, it often only reduces the set of actions the players could play
- We need another way to find what action each player will finally play!